Project Name:
Study on Fructose Intolerance on *Oryctolagus cuniculus*

General Objective:
How do modifications in the sequence and structure in the gene/protein describe the molecular basis of disease, and what can comparing these variations to the model structure tell us about the injured biological function?

Specific Objectives:

1. Find out what the cause of the disease is from a genetic, proteic, structural and functional perspective .
2. Set up the repository structure required to store and search curated datasets.
3. Provide capacitation to the lab team to ensure the tool's longevity.

Milestones:

M1: Research papers agreed
M2: Data migrated and validated
M3: Status completed
M4: Repository published

Project Members:

Jana Iturriaga Pizarro (Bioinformatics undergraduate student, Universitat Politècnica de Catalunya) - Principal investigator, secondary writer.

Júlia Grau Martín (Bioinformatics undergraduate student, Universitat Politècnica de Catalunya) - Project coordinator, secondary investigator, secondary writer.

Diana Zhou Chen (Bioinformatics undergraduate student, Universitat Politècnica de Catalunya) - Lead writer, secondary investigator, risk supervisor.

Josep Ferré Parés (Bioinformatics undergraduate student, Universitat Politècnica de Catalunya) - Communication and design manager, secondary investigator, secondary writer.

1. Break the project into Work Packages (WP)

WP1 Research papers agreed: This work package focuses on establishing the project foundation by consolidating scientific literature, setting clear project parameters, and defining success criteria for the dataset repository.

WP2 Data migrated and validated: Covers the technical implementation of the data repository infrastructure, migration of historical experimental data, and quality assurance processes.
WP3 Project Launch and handover: Ensures operational readiness of the repository through user documentation, team training, and formal transition to ongoing maintenance.

2. For each WP break down its specific Tasks (T)

WP1

1.1 Find Research papers on PubMed on Hereditary Fructose Intolerance

1.2 Define and Conceptualize the full structure of the Illness causing protein using PDB datasource

1.3 Find information about the mutations causing the HFI on *Oryctolagus cuniculus*

WP2

2.1 Create a plan to organize genetic and protein information

2.2 Research historical experimental data on *Oryctolagus cuniculus* fructose intolerance

2.3 Validate the schema of the structure using the sources found in WP1

2.4 Migrate data into GitHub repository and cross-reference with PDB/PubMed data researched WP1

2.5 Perform data integrity, metadata completeness, and functional annotation validation checks

WP3

3.1 Develop and organize user documentation, ensuring that all external sources, databases, software, tools, and guidelines used are properly cited according to the requirement

3.2 Review the database and documentation and verify that the repository is functional and accessible.

3.3 Publish the final version.

3. Plan risks while you plan the work

Score = Likelihood + Impact

Score Category
1–2 Low
3–5 Medium
6–10 High

Risk	Likelihood	Impact	Final Score	Category
Not finding enough recent research papers on HFI	3	5	8	High
Insufficient information about mutations related to HFI in rabbit	3	4	6	Medium
Missing data or references in the final repository	2	3	5	Medium
Having problems or error when migrating data to the Github	2	2	4	Medium
Not having enough time to finish the project	1	8	9	High

Our identified risk is not finding enough recent research papers on HFI, we have realized because there may not be many recent papers that are related to our project topic; our contingency consists of using less recent databases or using different keywords.

Our identified risk is having insufficient information about mutations related to HFI in rabbits; we have realized this because it may be a very specific topic. Our contingency plan consists of searching in other scientific databases and using information from related species if it's necessary.

Our identified risk is missing data or references in the final repository we have realized because the project uses information from different sources and webs; our contingency plan consists in writing down where our information comes from and comparing the data with the original paper.

Our identified risk is having problems or errors when migrating data to GitHub; we have realized this because it is our first time creating and managing a repository, so we are not completely familiar with the process. Our contingency plan consists of keeping a backup of the data and checking the files after uploading them to GitHub.

Our identified risk is not having enough time to finish the project, we have realized this because the projects has several different milestone, our contingency plan consists of prioritizing the most important tasks and setting deadlines for each WP.

Gantt Chart → Gantt Chart

Uniprot →

- https://www.uniprot.org/uniprotkb/P05062/entry (P05062 · ALDOB_HUMAN)
- https://www.uniprot.org/uniprotkb/P79226/entry (P79226 · ALDOB_RABIT (*Oryctolagus cuniculus*))